

# **HEART FAILURE 30-DAY READMISSION PREDICTION**

Machine Learning-Based Classification of Patients at Risk  
of 30-Day Readmission

**Project Documentation Report**

**Submitted as part of the 3Skill Internship**

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**Project:** Heart Failure Readmission Prediction

**Technology Used:** Python, Pandas, NumPy, Scikit-learn,  
Matplotlib, Seaborn

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## **ABSTRACT**

Heart failure is a major healthcare concern, and patients who have been hospitalized because of heart failure may be at risk of being readmitted within 30 days. Early identification of patients at higher risk of readmission can support better discharge planning, follow-up, and post-discharge care.

This project focuses on developing machine learning classification models to predict whether a patient will be readmitted within 30 days following a heart-failure-related hospitalization. The dataset contains 12,000 patient records with demographic, clinical, lifestyle, and hospitalization-related features. The target variable, `Readmitted_30_Days`, indicates whether a patient was readmitted within 30 days.

Three machine learning algorithms were developed and evaluated: Logistic Regression, K-Nearest Neighbors (KNN), and Decision Tree. The data was preprocessed using feature scaling for numerical variables and one-hot encoding for categorical variables. An 80:20 stratified train-test split was used to maintain the class distribution. Hyperparameter tuning was performed using cross-validation with ROC-AUC as the primary scoring metric.

Among the final tuned models, Logistic Regression achieved the highest test ROC-AUC of 0.9561, with an accuracy of 89.42%, precision of 85.20%, recall of 78.33%, and F1-score of 81.62%. KNN achieved a test ROC-AUC of 0.9271, while the Decision Tree achieved 0.8947. Based on the evaluated test-set metrics, Logistic Regression was selected as the final model for this project.

The results demonstrate that machine learning can be used to identify patterns associated with 30-day heart failure readmission. However, further external validation and clinical evaluation would be required before using such a model in a real healthcare environment.

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# 1. INTRODUCTION

failure is a serious cardiovascular condition that can require hospitalization and continued medical care. Patients who have been hospitalized for heart failure may have different levels of risk for returning to the hospital after discharge. Identifying patients who are more likely to experience readmission can help healthcare teams focus follow-up and post-discharge support on patients who may require greater attention.

Machine learning provides methods for analyzing patient data and identifying patterns that may be associated with clinical outcomes. In this project, machine learning classification techniques are applied to a heart failure readmission dataset to predict whether a patient will be readmitted within 30 days following hospitalization.

The project uses a dataset containing 12,000 patient records and includes demographic, clinical, lifestyle, and hospitalization-related features. The target variable is `Readmitted_30_Days`, which represents whether the patient experienced readmission within 30 days.

Three classification algorithms are developed and compared: Logistic Regression, K-Nearest Neighbors (KNN), and Decision Tree. The models are evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Hyperparameter tuning and cross-validation are also performed to improve model performance and assess the consistency of the models.

The overall objective of this project is to develop and evaluate machine learning models for identifying patients at risk of 30-day readmission and to determine how effectively different classification approaches perform on the given dataset.

## **2. PROBLEM STATEMENT & OBJECTIVES**

### **2.1 Problem Statement**

The hospital wants to identify patients who are at higher risk of 30-day readmission after a heart-failure-related hospitalization. Early identification of high-risk patients can help the care team prioritize follow-up, discharge planning, and post-discharge support.

The objective of this project is to use the provided heart-failure readmission dataset to develop and evaluate machine learning classification models that can predict whether a patient will be readmitted within 30 days. Three different binary classification algorithms are trained, tuned, and compared using appropriate evaluation metrics.

### **2.2 Objectives**

Use a numbered list:

1. To understand and explore the heart failure readmission dataset.
2. To perform data quality checks and preprocessing to prepare the dataset for machine learning.
3. To develop three binary classification models: Logistic Regression, K-Nearest Neighbors (KNN), and Decision Tree.
4. To evaluate the models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
5. To perform hyperparameter tuning using cross-validation and ROC-AUC as the primary scoring metric.
6. To compare the final tuned models based on their test-set performance.
7. To analyze model generalization by examining train-test performance and potential overfitting or underfitting.
8. To identify the final model based on the overall evaluation results while recognizing that further external and clinical validation would be required before real-world use.

### 3. DATASET DESCRIPTION

#### 3.1 Dataset Overview

The dataset used in this project contains 12,000 patient records related to heart failure. Each record represents a patient and contains demographic, clinical, lifestyle, and hospitalization-related information. The dataset is used to develop a binary classification model for predicting whether a patient will be readmitted within 30 days following hospitalization.

The target variable used for prediction is `Readmitted_30_Days`. A value of 0 represents that the patient was not readmitted within 30 days, while a value of 1 represents that the patient was readmitted within 30 days.

#### 3.2 Dataset Features

| Feature Category            | Examples                                                          |
|-----------------------------|-------------------------------------------------------------------|
| Patient Information         | Patient ID, Gender, Age                                           |
| Lifestyle Factors           | Smoking Status, Alcohol Consumption                               |
| Clinical Information        | Heart Failure Type, NYHA Class                                    |
| Hospitalization Information | Previous HF Admissions, ICU Admission                             |
| Other Patient Features      | Other demographic and clinical variables available in the dataset |
| Target Variable             | <code>Readmitted_30_Days</code>                                   |

#### 3.3 Target Variable

The target variable, `Readmitted_30_Days`, is a binary variable used to classify patients into two groups: patients who were not readmitted within 30 days and patients who were readmitted within 30 days. The dataset contains 8,400 records in class 0 and 3,600 records in class 1, corresponding to 70% and 30%, respectively. This indicates that the target variable is moderately imbalanced and should be considered during model development and evaluation.

## 4. DATA EXPLORATION / EDA

### 4.1 Initial Dataset Inspection

The dataset was initially inspected to understand its structure, dimensions, feature names, data types, and overall contents. The dataset contains 12,000 patient records. Initial inspection was performed using the first few records, dataset shape, column information, and missing-value checks.

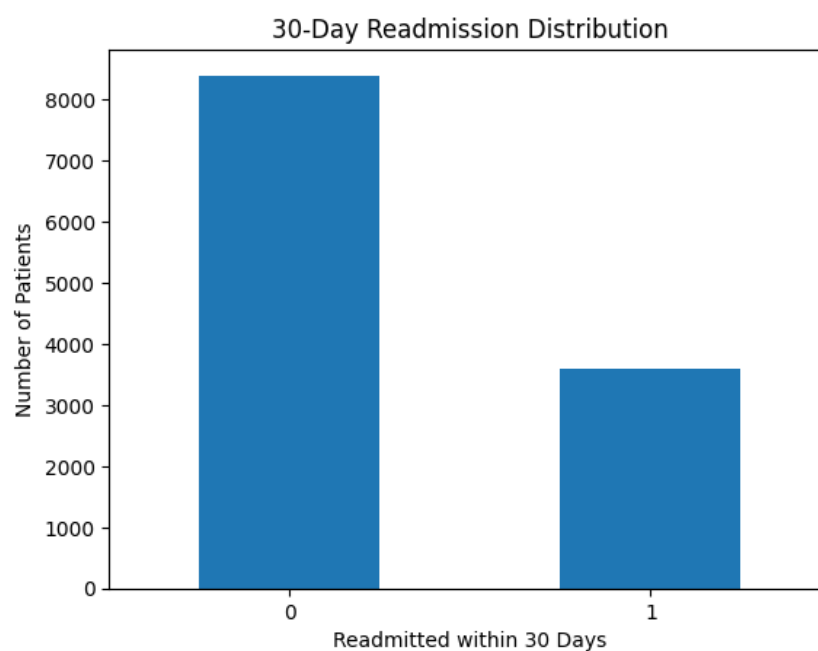
This initial analysis helped establish an understanding of the available variables and identify potential data quality issues before proceeding with preprocessing and model development.

### 4.2 Target Variable Distribution

The distribution of the target variable `Readmitted_30_Days` was analyzed to understand the proportion of patients who were and were not readmitted within 30 days.

The dataset contains 8,400 patients who were not readmitted (Class 0) and 3,600 patients who were readmitted (Class 1). Therefore, Class 0 represents 70% of the dataset, while Class 1 represents 30%.

This shows that the target variable is imbalanced, with fewer patients experiencing 30-day readmission. Therefore, evaluation metrics such as Precision, Recall, F1-Score, and ROC-AUC were considered along with Accuracy.



## 4.3 Descriptive Statistics

Descriptive statistical analysis was performed on the numerical variables to understand their central tendency, variability, and range. Measures such as mean, standard deviation, minimum, maximum, and quartiles were examined.

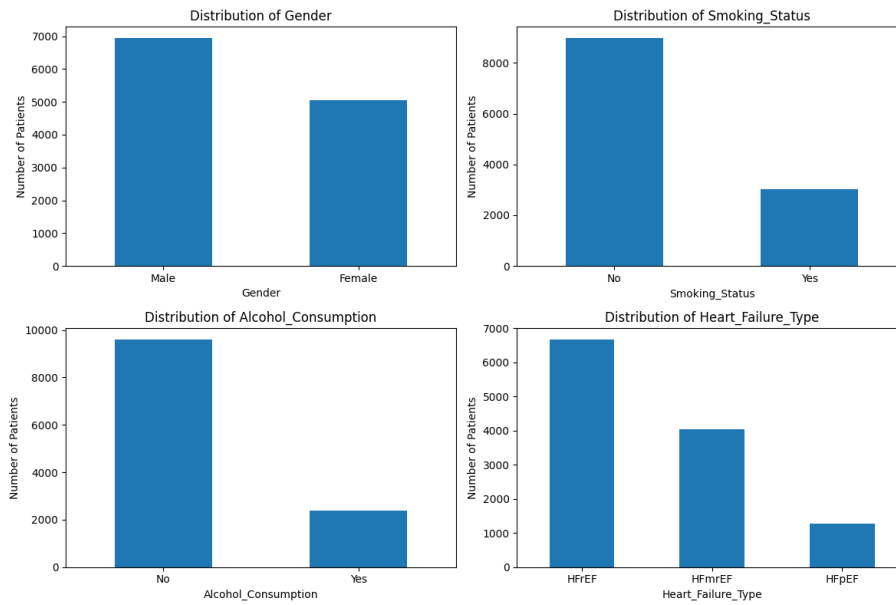
This analysis provided an overview of the numerical characteristics of the patient population and helped identify the general distribution and range of the numerical features before preprocessing.

|                              | count   | mean       | std        | min        | 25%        | 50%        | 75%        | max         |
|------------------------------|---------|------------|------------|------------|------------|------------|------------|-------------|
| Age                          | 12000.0 | 67.396417  | 13.079380  | 30.000000  | 60.000000  | 68.000000  | 77.000000  | 95.000000   |
| BMI                          | 12000.0 | 27.790903  | 5.035375   | 16.000000  | 24.299963  | 27.780840  | 31.246772  | 45.975566   |
| Exercise_Frequency           | 12000.0 | 1.018583   | 1.036819   | 0.000000   | 0.000000   | 1.000000   | 2.000000   | 4.000000    |
| Hypertension                 | 12000.0 | 0.412417   | 0.492290   | 0.000000   | 0.000000   | 0.000000   | 1.000000   | 1.000000    |
| Diabetes                     | 12000.0 | 0.296667   | 0.456807   | 0.000000   | 0.000000   | 0.000000   | 1.000000   | 1.000000    |
| Chronic_Kidney_Disease       | 12000.0 | 0.268000   | 0.442936   | 0.000000   | 0.000000   | 0.000000   | 1.000000   | 1.000000    |
| Coronary_Artery_Disease      | 12000.0 | 0.401500   | 0.490222   | 0.000000   | 0.000000   | 0.000000   | 1.000000   | 1.000000    |
| Previous_Stroke              | 12000.0 | 0.183083   | 0.386751   | 0.000000   | 0.000000   | 0.000000   | 0.000000   | 1.000000    |
| Atrial_Fibrillation          | 12000.0 | 0.277333   | 0.447701   | 0.000000   | 0.000000   | 0.000000   | 1.000000   | 1.000000    |
| Previous_HF_Admissions       | 12000.0 | 0.631083   | 0.874765   | 0.000000   | 0.000000   | 0.000000   | 1.000000   | 7.000000    |
| Previous_Hospital_Admissions | 12000.0 | 1.694083   | 1.369241   | 0.000000   | 1.000000   | 1.000000   | 2.000000   | 9.000000    |
| NYHA_Class                   | 12000.0 | 2.273667   | 1.020307   | 1.000000   | 1.000000   | 2.000000   | 3.000000   | 4.000000    |
| Ejection_Fraction            | 12000.0 | 38.644836  | 9.193389   | 10.000000  | 32.675299  | 38.697234  | 44.728884  | 71.874767   |
| Systolic_BP                  | 12000.0 | 124.711841 | 18.578098  | 85.000000  | 111.789513 | 124.570115 | 137.288621 | 194.855837  |
| Diastolic_BP                 | 12000.0 | 77.983720  | 12.043780  | 50.000000  | 69.758189  | 77.959766  | 86.250475  | 120.000000  |
| Heart_Rate                   | 12000.0 | 83.821400  | 12.366242  | 50.000000  | 75.284337  | 83.791648  | 92.242255  | 132.041953  |
| Oxygen_Saturation            | 12000.0 | 94.260086  | 2.590572   | 84.502044  | 92.479819  | 94.272651  | 96.061874  | 100.000000  |
| Creatinine                   | 12000.0 | 1.881805   | 1.850859   | 0.500000   | 0.500000   | 1.150088   | 2.411102   | 8.000000    |
| Sodium                       | 12000.0 | 138.457719 | 3.568943   | 124.373997 | 136.079395 | 138.462740 | 140.876584 | 150.000000  |
| Potassium                    | 12000.0 | 4.308370   | 0.530434   | 2.800000   | 3.949353   | 4.301733   | 4.658113   | 6.418538    |
| Hemoglobin                   | 12000.0 | 12.177487  | 1.946681   | 7.000000   | 10.858640  | 12.201073  | 13.525338  | 18.000000   |
| Blood_Glucose                | 12000.0 | 118.077890 | 32.696853  | 70.000000  | 92.609403  | 115.999968 | 139.825765 | 258.524090  |
| BNP                          | 12000.0 | 204.936134 | 203.739039 | 50.000000  | 84.818008  | 142.847005 | 248.264021 | 4097.436155 |
| Length_of_Stay               | 12000.0 | 8.146667   | 6.472865   | 1.000000   | 3.000000   | 6.000000   | 11.000000  | 30.000000   |
| ICU_Admission                | 12000.0 | 0.339917   | 0.473700   | 0.000000   | 0.000000   | 0.000000   | 1.000000   | 1.000000    |
| Emergency_Admission          | 12000.0 | 0.511667   | 0.499885   | 0.000000   | 0.000000   | 1.000000   | 1.000000   | 1.000000    |
| Beta_Blocker                 | 12000.0 | 0.804167   | 0.396857   | 0.000000   | 1.000000   | 1.000000   | 1.000000   | 1.000000    |
| ACE_ARB                      | 12000.0 | 0.762000   | 0.425877   | 0.000000   | 1.000000   | 1.000000   | 1.000000   | 1.000000    |
| Diuretic                     | 12000.0 | 0.576667   | 0.494108   | 0.000000   | 0.000000   | 1.000000   | 1.000000   | 1.000000    |
| SGLT2_Inhibitor              | 12000.0 | 0.395000   | 0.488871   | 0.000000   | 0.000000   | 0.000000   | 1.000000   | 1.000000    |
| Readmitted_30_Days           | 12000.0 | 0.300000   | 0.458277   | 0.000000   | 0.000000   | 0.000000   | 1.000000   | 1.000000    |

## 4.4 Categorical Feature Analysis

The categorical variables were analyzed using frequency counts to understand the distribution of different patient characteristics. Features such as Gender, Smoking Status, Alcohol Consumption, and Heart Failure Type were examined.

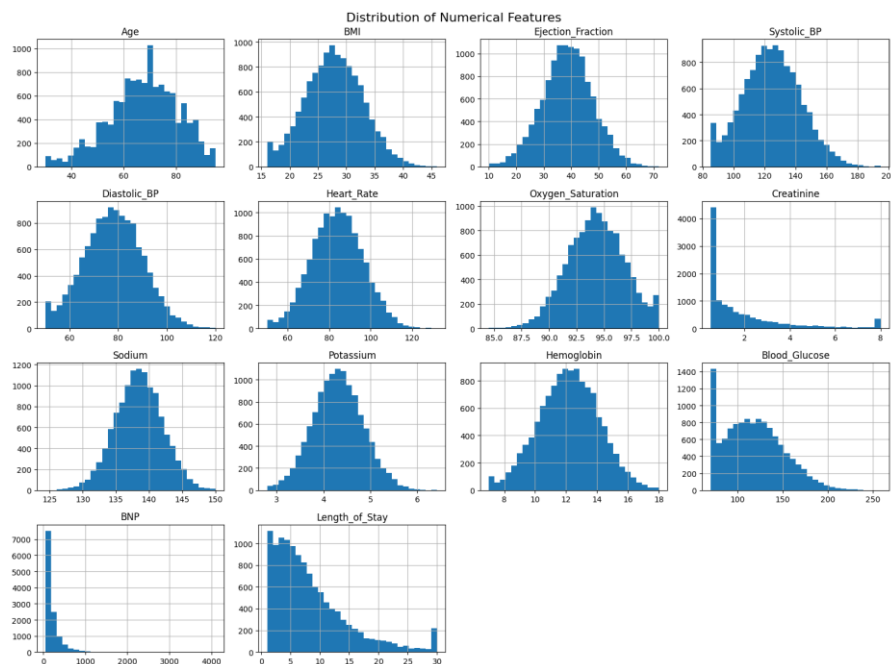
Bar charts were used to visualize the frequency of different categories and identify the distribution of patients across these groups.



## 4.5 Numerical Feature Distributions

Histograms were used to examine the distributions of numerical features in the dataset. This helped identify the range and general distribution of patient-related numerical variables and provided useful information for subsequent preprocessing.

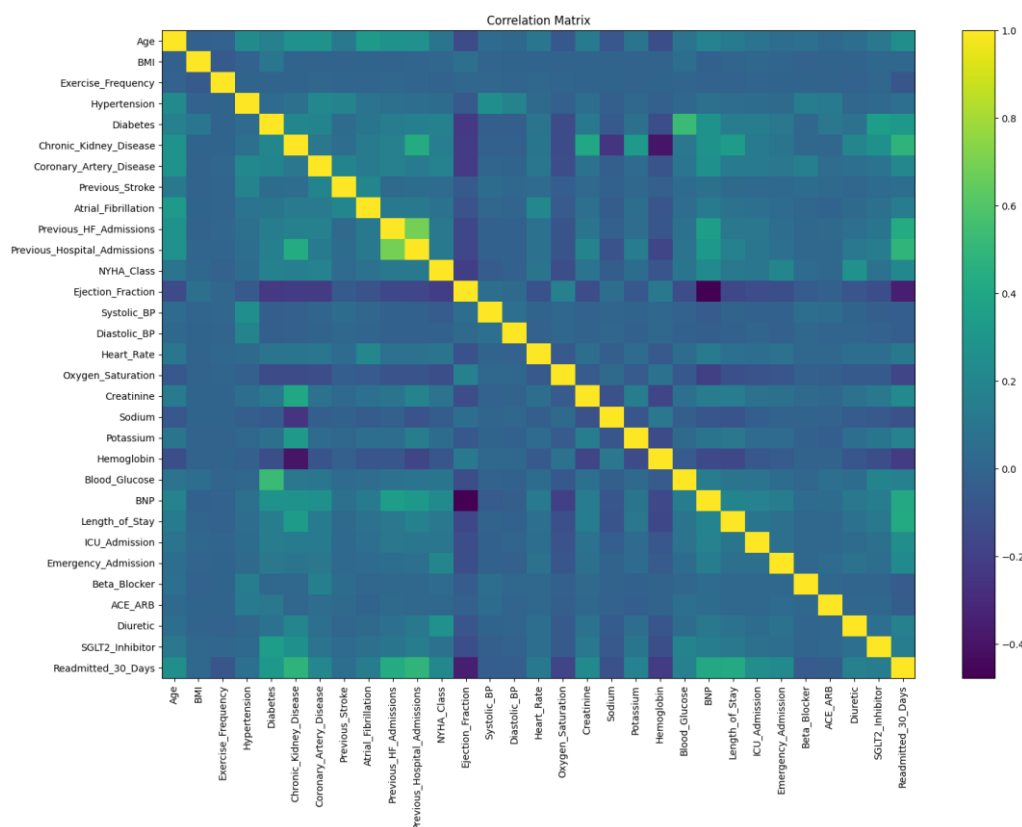
The distributions were reviewed before applying standardization to the numerical features during the preprocessing stage.



## 4.6 Correlation Analysis

A correlation matrix was generated to examine relationships between numerical features. Correlation analysis helps identify the strength and direction of linear relationships between variables.

The correlation matrix was also used to examine the relationship between the available numerical features and the target variable. This provided an initial understanding of which variables showed stronger or weaker associations with 30-day readmission.



## 4.7 Target Relationship Analysis

The relationship between selected patient and hospitalization characteristics and 30-day readmission was further explored. Readmission rates were examined across variables such as Heart Failure Type, NYHA Class, Previous Heart Failure Admissions, and ICU Admission.

Mean numerical feature values were also compared between patients who were readmitted and those who were not. These analyses helped identify patterns in the dataset that could be relevant to the prediction task.

|                                         |       |          |                                                        |            |            |
|-----------------------------------------|-------|----------|--------------------------------------------------------|------------|------------|
| Readmission Rate by Heart Failure Type: |       |          | Readmission Rate by Previous Heart Failure Admissions: |            |            |
|                                         | count | mean     |                                                        | count      | mean       |
| Heart_Failure_Type                      |       |          | Previous_HF_Admissions                                 |            |            |
| HFrEF                                   | 6679  | 0.411738 | 0                                                      | 6785       | 0.154753   |
| HFmrEF                                  | 4044  | 0.183976 | 1                                                      | 3503       | 0.381958   |
| HFpEF                                   | 1277  | 0.083007 | 2                                                      | 1221       | 0.640459   |
|                                         |       |          | 3                                                      | 368        | 0.853261   |
|                                         |       |          | 4                                                      | 93         | 0.924731   |
|                                         |       |          | 5                                                      | 29         | 1.000000   |
|                                         |       |          | 7                                                      | 1          | 1.000000   |
| Readmission Rate by NYHA Class:         |       |          | Numerical Feature Comparison by Readmission:           |            |            |
|                                         | count | mean     | Readmitted_30_Days                                     | 0          | 1          |
| NYHA_Class                              |       |          | Age                                                    | 65.234286  | 72.441389  |
| 1                                       | 3598  | 0.145914 | BMI                                                    | 27.740206  | 27.909195  |
| 2                                       | 2975  | 0.338487 | Ejection_Fraction                                      | 40.778738  | 33.665732  |
| 3                                       | 3972  | 0.356999 | Systolic_BP                                            | 125.170573 | 123.641466 |
| 4                                       | 1455  | 0.446735 | Heart_Rate                                             | 82.849194  | 86.089879  |
|                                         |       |          | Creatinine                                             | 1.613329   | 2.508248   |
|                                         |       |          | Sodium                                                 | 138.715328 | 137.856631 |
|                                         |       |          | Hemoglobin                                             | 12.461187  | 11.515519  |
|                                         |       |          | Blood_Glucose                                          | 114.461252 | 126.516712 |
|                                         |       |          | BNP                                                    | 149.825371 | 333.527914 |
|                                         |       |          | Length_of_Stay                                         | 6.350476   | 12.337778  |
| Readmission Rate by ICU Admission:      |       |          |                                                        |            |            |
|                                         | count | mean     |                                                        |            |            |
| ICU_Admission                           |       |          |                                                        |            |            |
| 0                                       | 7921  | 0.216134 |                                                        |            |            |
| 1                                       | 4079  | 0.462859 |                                                        |            |            |

## 5. DATA QUALITY & PREPROCESSING

### 5.1 Missing Value Analysis

The dataset was checked for missing values to ensure that incomplete records would not negatively affect the analysis or machine learning models. The missing-value analysis was performed for all features in the dataset. No problematic missing values requiring imputation were identified.

### 5.2 Duplicate Record Analysis

Duplicate records were checked to ensure that repeated observations did not affect the analysis or introduce bias into the machine learning models. The dataset was examined for completely duplicated rows as part of the data quality assessment.

### 5.3 Duplicate Patient ID Analysis

The Patient\_ID feature was checked for duplicate values to determine whether multiple records were associated with the same patient identifier. This check helped assess the uniqueness of patient records.

## **5.4 Binary Feature Validation**

Binary features were examined to verify that they contained valid binary values. This validation helped identify any unexpected values that could cause problems during preprocessing or model training.

## **5.5 Categorical Feature Validation**

Categorical features were checked to ensure that their values were consistent with the expected categories. This helped ensure that the categorical variables could be appropriately encoded before model training.

## **5.6 Numerical Range Validation**

Numerical features were examined to identify values outside their expected ranges. This step helped detect potentially invalid or unrealistic observations before the data was used for machine learning.

## **5.7 Data Leakage Analysis**

The features were reviewed for potential data leakage. Particular attention was given to ensuring that the target variable and information that would not be available at the time of prediction were not used as model inputs.

### **5.8 Feature and Target Separation**

The target variable `Readmitted_30_Days` was separated from the input features. The `Patient_ID` column was excluded because it is an identifier and does not represent a meaningful predictive characteristic.

The remaining features were divided into numerical and categorical variables for preprocessing.

## **5.9 Train-Test Split**

The dataset was divided into training and testing sets using an 80:20 stratified split. Stratification was used to preserve the proportion of the two target classes in both subsets. A `random_state` of 42 was used to ensure reproducibility.

The training data was used for model training and hyperparameter tuning, while the test data was kept separate for final evaluation.

```
Training data: (9600, 34)
Testing data: (2400, 34)

Training target distribution:
Readmitted_30_Days
0    0.7
1    0.3
Name: proportion, dtype: float64

Testing target distribution:
Readmitted_30_Days
0    0.7
1    0.3
Name: proportion, dtype: float64
```

## 5.10 Feature Preprocessing

Numerical features were standardized using StandardScaler. Categorical features were transformed using One-Hot Encoding with handle\_unknown="ignore".

A ColumnTransformer was used to apply the appropriate preprocessing technique to each feature type. The preprocessing pipeline was fitted on the training data and then used to transform both the training and test data. This prevented information from the test set from being used during the preprocessing stage.

# 6. MACHINE LEARNING MODELS

## 6.1 Model Selection

Three supervised machine learning classification algorithms were selected for the 30-day heart failure readmission prediction task: Logistic Regression, K-Nearest Neighbors (KNN), and Decision Tree.

These models represent different classification approaches. Logistic Regression provides a linear classification method, KNN makes predictions based on neighboring observations, and Decision Tree uses a tree-based structure to make classification decisions.

The three models were trained using the preprocessed training data and evaluated on the held-out test data. The models were compared using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

## **6.2 Logistic Regression**

Logistic Regression was used as one of the baseline classification models. It estimates the probability of a binary outcome based on the input features. In this project, it was used to estimate the probability that a patient would be readmitted within 30 days.

Logistic Regression provides a relatively interpretable classification approach and was later tuned using cross-validation to identify suitable hyperparameters.

## **6.3 K-Nearest Neighbors (KNN)**

K-Nearest Neighbors (KNN) is a classification algorithm that predicts the class of an observation based on the classes of nearby observations. The number of neighboring observations considered is controlled by the `n_neighbors` parameter.

KNN was included to provide a non-parametric approach for comparison with Logistic Regression and Decision Tree. Feature scaling was applied before using KNN because the algorithm relies on distances between observations.

## **6.4 Decision Tree**

Decision Tree was used as a tree-based classification algorithm. It makes predictions by recursively splitting the data according to feature conditions until reaching decision nodes and leaf nodes.

The Decision Tree model can capture non-linear relationships between features and the target variable. The `max_depth` parameter was later tuned to control the complexity of the tree.

## **6.5 Evaluation Metrics**

The models were evaluated using multiple classification metrics to provide a comprehensive assessment of their performance.

Accuracy measures the proportion of correctly classified observations among all observations.

Precision measures the proportion of predicted positive cases that were actually positive.

Recall measures the proportion of actual positive cases that were correctly identified by the model.

F1-Score is the harmonic mean of Precision and Recall and provides a balance between the two metrics.

ROC-AUC measures the model's ability to distinguish between the two classes across different classification thresholds. ROC-AUC was also used as the primary scoring metric during hyperparameter tuning.

## **7. HYPERPARAMETER TUNING**

### **7.1 Purpose of Hyperparameter Tuning**

Hyperparameter tuning was performed to improve the performance of the machine learning models by identifying suitable model configurations. The tuning process was carried out using cross-validation, with ROC-AUC selected as the primary scoring metric.

The training dataset was used for hyperparameter tuning, while the test dataset was kept separate and was not used during the tuning process. This helped provide an unbiased evaluation of the final models on unseen data.

### **7.2 Logistic Regression Tuning**

Logistic Regression was tuned using GridSearchCV with 5-fold cross-validation. The hyperparameters considered included the regularization parameter C and the solver.

The best configuration obtained was:

- C: 0.1
- Solver: liblinear
- Maximum Iterations: 1000
- Random State: 42

The tuned Logistic Regression model achieved a mean cross-validation ROC-AUC of 0.9561 with a standard deviation of 0.0041.

### **7.3 K-Nearest Neighbors Tuning**

For K-Nearest Neighbors, different values of the `n_neighbors` parameter were evaluated using cross-validation. The tested values ranged from 3 to 31.

The best-performing configuration was:

- Number of Neighbors (K): 31

The tuned KNN model achieved a mean cross-validation ROC-AUC of 0.9271.

### **7.4 Decision Tree Tuning**

The Decision Tree model was tuned by evaluating different values of the `max_depth` parameter. The tested depths included 2, 3, 4, 5, 6, 7, 8, 10, 12, 15, and 20.

The best configuration was:

- Maximum Depth (`max_depth`): 6
- Random State: 42

The tuned Decision Tree achieved a mean cross-validation ROC-AUC of 0.8947.

### **7.5 Cross-Validation Results**

Five-fold cross-validation was used to assess the consistency of model performance across different subsets of the training data. The tuned Logistic Regression model produced ROC-AUC scores of 0.9514, 0.9562, 0.9501, 0.9450, and 0.9558 across the five folds.

The resulting mean ROC-AUC was 0.9561, with a standard deviation of 0.0041. The relatively small variation across the folds indicates that the model produced consistent ROC-AUC performance across the different validation subsets.

| <b>Fold</b>               | <b>ROC-AUC</b> |
|---------------------------|----------------|
| 1                         | 0.9514         |
| 2                         | 0.9562         |
| 3                         | 0.9501         |
| 4                         | 0.9450         |
| 5                         | 0.9558         |
| <b>Mean</b>               | 0.9517         |
| <b>Standard Deviation</b> | 0.0041         |

## 8. MODEL EVALUATION

### 8.1 Final Model Evaluation

The final tuned versions of Logistic Regression, K-Nearest Neighbors (KNN), and Decision Tree were evaluated using the unseen test dataset. The evaluation was performed using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

The test dataset was not used during hyperparameter tuning and was therefore used to provide an independent evaluation of the final model performance. The results provide information about how well each tuned model generalizes to unseen patient records.

### 8.2 Logistic Regression Evaluation

The tuned Logistic Regression model was evaluated on the test dataset using the selected hyperparameters. The model achieved an Accuracy of 89.42%, Precision of 85.20%, Recall of 78.33%, F1-Score of 81.62%, and ROC-AUC of 0.9561.

These results show the performance of the Logistic Regression model across the selected evaluation metrics.

| <b>Metric</b> | <b>Logistic Regression</b> |
|---------------|----------------------------|
| Accuracy      | 89.42%                     |
| Precision     | 85.20%                     |
| Recall        | 78.33%                     |
| F1-Score      | 81.62%                     |
| ROC-AUC       | 0.9561                     |

### 8.3 K-Nearest Neighbors Evaluation

The tuned K-Nearest Neighbors model was evaluated on the test dataset. The model achieved an Accuracy of 84.96%, Precision of 88.11%, Recall of 57.64%, F1-Score of 69.69%, and ROC-AUC of 0.9271.

These results provide the test-set performance of the tuned KNN model across the selected evaluation metrics.

| Metric    | KNN    |
|-----------|--------|
| Accuracy  | 84.96% |
| Precision | 88.11% |
| Recall    | 57.64% |
| F1-Score  | 69.69% |
| ROC-AUC   | 0.9271 |

### 8.4 Decision Tree Evaluation

The tuned Decision Tree model was evaluated on the test dataset. The model achieved an Accuracy of 84.50%, Precision of 78.62%, Recall of 66.39%, F1-Score of 71.99%, and ROC-AUC of 0.8947.

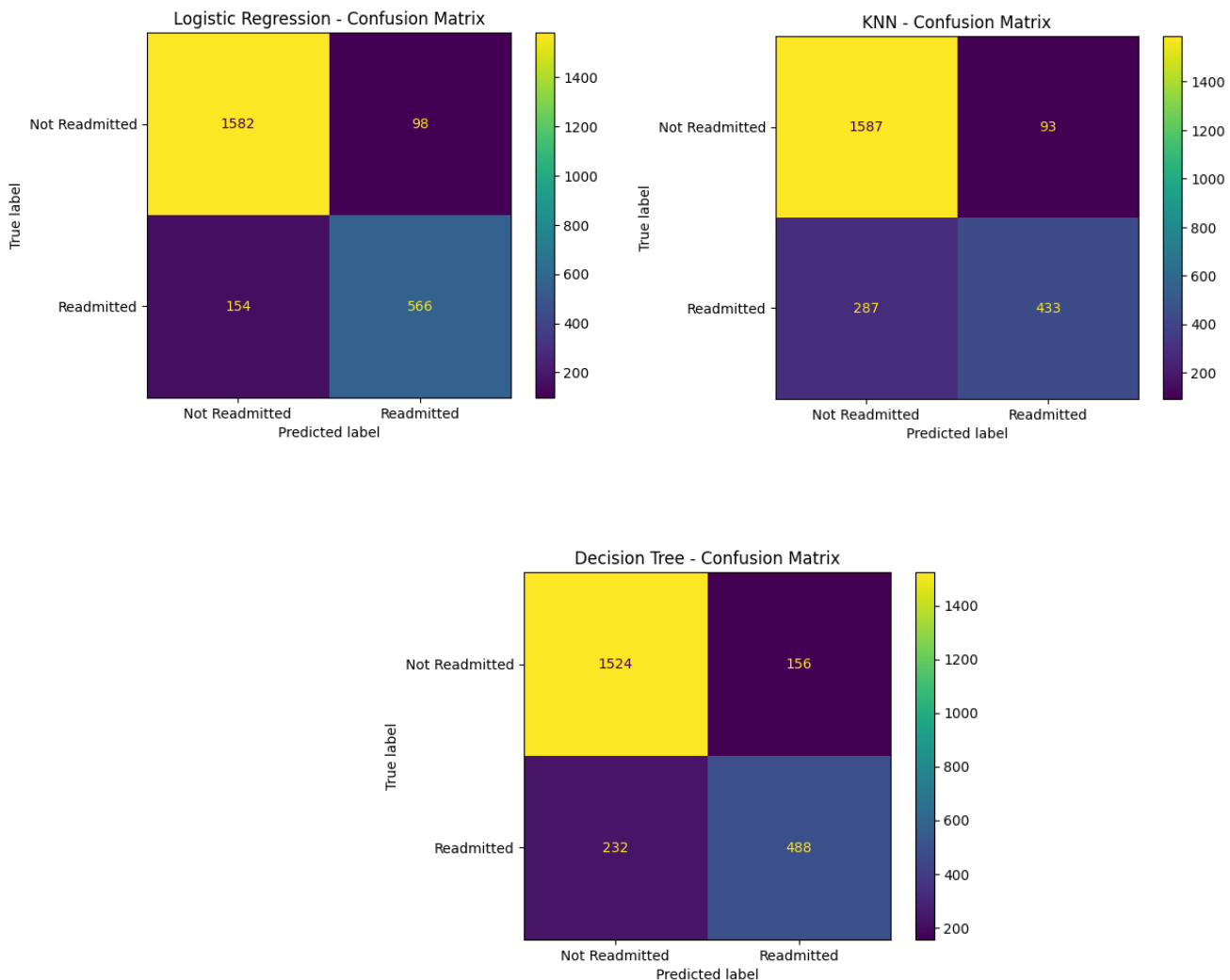
These results provide the test-set performance of the tuned Decision Tree model across the selected evaluation metrics.

| Metric    | Decision Tree |
|-----------|---------------|
| Accuracy  | 84.50%        |
| Precision | 78.62%        |
| Recall    | 66.39%        |
| F1-Score  | 71.99%        |
| ROC-AUC   | 0.8947        |

### 8.5 Confusion Matrix Analysis

Confusion matrices were generated for the three final tuned models to examine their classification results in terms of true positives, true negatives, false positives, and false negatives.

The confusion matrices provide a detailed view of the types of classification errors made by each model and complement the numerical evaluation metrics.

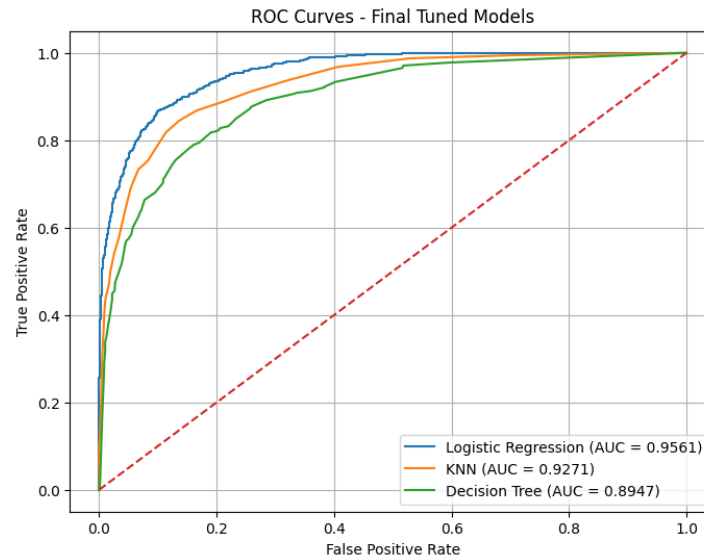


## 8.6 ROC Curve Analysis

ROC curves were used to visualize the classification performance of the three final tuned models across different classification thresholds.

The ROC-AUC value summarizes the ability of each model to distinguish between patients who were readmitted within 30 days and those who were not.

The ROC curve comparison was generated using the predicted probabilities from the final tuned models on the test dataset.



## 9. MODEL COMPARISON

### 9.1 Performance Comparison

The final tuned Logistic Regression, K-Nearest Neighbors (KNN), and Decision Tree models were compared using the evaluation metrics obtained from the test dataset. The comparison includes Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

The results are summarized in the following table.

| Model               | Accuracy | Precision | Recall | F1-Score | ROC-AUC |
|---------------------|----------|-----------|--------|----------|---------|
| Logistic Regression | 89.42%   | 85.20%    | 78.33% | 81.62%   | 0.9561  |
| KNN                 | 84.96%   | 88.11%    | 57.64% | 69.69%   | 0.9271  |
| Decision Tree       | 84.50%   | 78.62%    | 66.39% | 71.99%   | 0.8947  |

### 9.2 Comparison of Evaluation Metrics

The three models showed different performance across the evaluation metrics. Logistic Regression achieved an accuracy of 89.42% and a ROC-AUC of 0.9561 on the test dataset.

KNN achieved an accuracy of 84.96% and a ROC-AUC of 0.9271, while Decision Tree achieved an accuracy of 84.50% and a ROC-AUC of 0.8947.

The Precision, Recall, and F1-Score values also differed between the models. KNN recorded a Precision of 88.11%, while Logistic Regression recorded 85.20% and Decision Tree recorded 78.62%. The Recall values were 78.33% for Logistic Regression, 57.64% for KNN, and 66.39% for Decision Tree.

These differences demonstrate that the choice of evaluation metric can affect how model performance is interpreted. Therefore, multiple metrics were considered rather than relying on Accuracy alone.

### **9.3 ROC-AUC Comparison**

ROC-AUC was used as an important metric for comparing the classification performance of the three models. The final test ROC-AUC values were 0.9561 for Logistic Regression, 0.9271 for KNN, and 0.8947 for Decision Tree. The ROC curves generated in the previous section provide a visual comparison of the models across different classification thresholds.

### **9.4 Final Model Selection**

Based on the overall test-set evaluation, Logistic Regression was selected as the final model for this project. It achieved an Accuracy of 89.42%, Precision of 85.20%, Recall of 78.33%, F1-Score of 81.62%, and ROC-AUC of 0.9561.

The selection was based on the overall set of evaluation results rather than a single metric. The selected model is intended for this project and dataset; further external validation and clinical evaluation would be required before considering deployment in a real healthcare environment.

## **10.OVERFITTING / UNDERFITTING ANALYSIS**

### **10.1 Train-Test ROC-AUC Comparison**

To assess model generalization, the ROC-AUC performance of each final tuned model was compared between the training and test datasets.

Comparing training and test performance helps identify whether a model may be learning the training data too closely or failing to capture useful patterns.

| <b>Model</b>        | <b>Train<br/>ROC-AUC</b> | <b>Test<br/>ROC-AUC</b> | <b>Absolute Gap</b> |
|---------------------|--------------------------|-------------------------|---------------------|
| Logistic Regression | 0.9537                   | 0.9561                  | 0.0025              |
| KNN                 | 0.9398                   | 0.9271                  | 0.0127              |
| Decision Tree       | 0.9170                   | 0.8947                  | 0.0223              |

## 10.2 Logistic Regression

Logistic Regression achieved a training ROC-AUC of 0.9537 and a test ROC-AUC of 0.9561. The absolute difference between the two values is approximately 0.0025. The small difference indicates that the model's performance was consistent between the training and test datasets.

## 10.3 K-Nearest Neighbors

KNN achieved a training ROC-AUC of 0.9398 and a test ROC-AUC of 0.9271, resulting in an absolute difference of approximately 0.0127. The relatively small difference indicates that the model maintained similar performance on the training and test datasets, with a modest reduction in ROC-AUC on the test data.

## 10.4 Decision Tree

The Decision Tree achieved a training ROC-AUC of 0.9170 and a test ROC-AUC of 0.8947. The absolute difference was approximately 0.0223. Although the training performance was higher than the test performance, the difference was relatively small compared with a large train-test performance gap.

## 10.5 Overall Generalization Analysis

Overall, the train-test ROC-AUC comparison shows relatively small differences for all three final tuned models. Logistic Regression had the smallest absolute gap, followed by KNN and Decision Tree.

Based on these train-test ROC-AUC results, there is no strong evidence of severe overfitting or underfitting in the final tuned models. The models demonstrated reasonably consistent performance between the training and unseen test datasets.

## **11. LIMITATIONS**

### **11.1 Dataset Generalization**

The dataset used in this project contains 12,000 patient records. The results obtained from the dataset may not necessarily generalize to patients from different hospitals, healthcare systems, geographic regions, or patient populations. Additional external validation using independent datasets would be required to assess the generalizability of the models.

### **11.2 Class Imbalance**

The target variable contains 70% non-readmitted patients and 30% readmitted patients. This class imbalance was considered during model evaluation by using multiple metrics, including Precision, Recall, F1-Score, and ROC-AUC, rather than relying only on Accuracy.

### **11.3 Feature Availability and Data Leakage Risk**

The performance of a prediction model depends on the quality and availability of the input features. In a real healthcare environment, some variables may not be available at the exact time when a prediction is required. Therefore, careful assessment of feature availability and potential data leakage would be necessary before real-world implementation.

### **11.4 Model Assumptions and Complexity**

The three algorithms used in this project have different characteristics and assumptions. Logistic Regression provides a relatively simple linear approach, KNN relies on distances between observations, and Decision Tree uses a tree-based decision structure. These models may not capture every possible relationship present in a real-world healthcare dataset.

### **11.5 Clinical Validation**

The models were evaluated using machine learning performance metrics on the available dataset. The project does not include prospective clinical testing, external clinical validation, or evaluation of how predictions would affect patient outcomes. Therefore, the models should not be considered clinically validated solely based on the results obtained in this project.

## **11.6 Prediction Threshold**

The classification results depend on the decision threshold used to convert predicted probabilities into class predictions. In a real healthcare application, the appropriate threshold would need to be determined according to the intended clinical use, the consequences of false positives and false negatives, and clinical requirements.

## **12. CONCLUSION**

This project focused on developing machine learning models to predict 30-day readmission among patients following a heart-failure-related hospitalization. The dataset contained 12,000 patient records with demographic, clinical, lifestyle, and hospitalization-related features.

The dataset was explored and examined for data quality issues before model development. Data preprocessing included separating numerical and categorical features, standardizing numerical variables, applying one-hot encoding to categorical variables, and using an 80:20 stratified train-test split. These steps helped prepare the data for reliable model training and evaluation.

Three classification algorithms were developed and evaluated: Logistic Regression, K-Nearest Neighbors (KNN), and Decision Tree. Hyperparameter tuning was performed using cross-validation with ROC-AUC as the primary scoring metric. The final tuned models were then evaluated on the unseen test dataset using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

The final test results showed that Logistic Regression achieved an Accuracy of 89.42%, Precision of 85.20%, Recall of 78.33%, F1-Score of 81.62%, and ROC-AUC of 0.9561. KNN achieved a ROC-AUC of 0.9271, while Decision Tree achieved a ROC-AUC of 0.8947.

Based on the overall test-set evaluation, Logistic Regression was selected as the final model for this project. The train-test ROC-AUC comparison also showed relatively small performance gaps across the three models, indicating reasonably consistent generalization on the available dataset.

Overall, the project demonstrates how machine learning classification techniques can be applied to a heart failure readmission prediction problem. However, the results are based on the available dataset and should not be interpreted as clinical validation. External validation, prospective evaluation, and clinical assessment would be necessary before applying such a model in a real healthcare environment.

## **13. REFERENCES**

### **13.1 Software and Libraries**

The following software libraries and documentation were used during the development of this project:

1. Python – Programming language used for data analysis and machine learning implementation.
2. Pandas – Used for data loading, manipulation, exploration, and preprocessing.
3. NumPy – Used for numerical operations and data processing.
4. Scikit-learn – Used for data preprocessing, machine learning models, hyperparameter tuning, cross-validation, and evaluation metrics.
5. Matplotlib – Used for creating data visualizations and model evaluation plots.
6. Seaborn – Used for statistical visualizations, including correlation heatmaps and other exploratory data analysis plots.

### **13.2 Dataset**

The primary dataset used in this project was the provided heart failure readmission dataset containing 12,000 patient records. The dataset was supplied as part of the project/internship resources and was used for exploratory analysis, preprocessing, model development, and evaluation.